A Leakage-Safe Temporal-Evidence Pipeline for Assessing Network Impact on Manufacturing Efficiency in Smart Factories: A Case Study

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***Abstract*—The paradigm of Industry 5.0 and smart manufacturing envisions pervasive integration of 6G Ultra-Reliable Low-Latency Communication (URLLC) to drive real-time closed-loop control and autonomous cyber-physical coordination. However, applied machine learning (ML) models evaluating network telemetry frequently suffer from methodological vulnerabilities, including temporal data leakage, target contamination, and unvalidated synthetic benchmark assumptions. In this paper, we propose an end-to-end, leakage-safe temporal-evidence pipeline for auditing network telemetry and quantifying its causal and predictive influence on industrial manufacturing efficiency. Through comprehensive Kolmogorov-Smirnov (KS) goodness-of-fit testing across nine industrial features, we reveal that the studied enterprise telemetry dataset ($N = 100,000$ observations across 50 machines over 68.4 days) is statistically consistent with independent uniform generation; no pairwise feature correlation survives Bonferroni correction across all 36 pairs (maximum $|r| = 0.0075$ between Vibration and Network Latency, uncorrected $p = 0.0170$, adjusted $p = 0.6127$). While a naive segmented regression search suggests a latency breakpoint at 20.7 ms ($F = 4.3127$, naive asymptotic $p = 0.0381$), a rigorous supremum-F permutation test ($N = 2,000$, seed=42) reveals $p = 0.1640$ (FDR $q = 0.5100$), establishing the threshold as exploratory rather than confirmatory. Utilizing leak-free temporal and machine-grouped validation splits, we benchmark Random Forest and LightGBM models against naive baselines: network telemetry achieves near-chance classification discrimination (ROC-AUC 0.4949–0.5073) for composite efficiency, failing to outperform random guessing. Our findings highlight the critical necessity of rigorous data audits and leak-free temporal benchmarking in smart factory research.**

***Index Terms*—Industrial IoT, 6G URLLC, Predictive Maintenance, Temporal Leakage, Statistical Audit, Changepoint Detection, Machine Learning in Manufacturing.**

I. Introduction

A. The Promise: 6G URLLC in Industry 4.0/5.0

The convergence of advanced telecommunications and industrial automation forms the core infrastructure of Industry 4.0 and 5.0. Future Sixth-Generation (6G) wireless communication networks promise to deliver sub-millisecond round-trip latencies, extreme jitter bounds, and block reliability rates exceeding 99.99999% (10^{-7} block error rate) [1], [2]. In mission-critical cyber-physical production systems (CPPS), such Ultra-Reliable Low-Latency Communication (URLLC) capabilities are designed to underpin closed-loop robotic cell collaboration, distributed programmable logic controllers (PLCs), edge-native computer vision, and real-time automated guided vehicle (AGV) routing [3], [4].

Standardization bodies, notably 3GPP in Technical Specification TS 22.104 [5], have articulated precise communication service requirements for vertical domains, specifying end-to-end latency tolerances between 0.5 ms and 20 ms for motion control and machine interaction. In parallel, smart factory operators increasingly ingest wireless telemetry (e.g., latency, jitter, packet loss) alongside operational technology (OT) machine sensors (e.g., temperature, vibration, power consumption) into predictive artificial intelligence pipelines to forecast production throughput and avoid costly operational downtime.

B. The Problem: Data Leakage and Optimistic Evaluation

Despite significant investments in smart manufacturing analytics, a profound reproducibility crisis looms over applied predictive modeling [6]. Recent methodological surveys demonstrate that a dominant fraction of published smart factory and IoT machine learning architectures report inflated classification accuracies due to subtle forms of data leakage [7]. Specifically, two vulnerabilities are rampant:

1) **Temporal Autocorrelation and Group Contamination**: Standard random K-fold cross-validation mixes temporally adjacent observations across train and test partitions. When machines operate under continuous autoregressive processes, random splits leak temporal dependencies, causing models to memorize specific time-series trajectories rather than generalizing to unseen temporal horizons [8].
2) **Target Contamination / Proxy Feature Leakage**: Target labels such as "Overall Equipment Efficiency" (OEE) or categorical "Operational Efficiency" are frequently derived as deterministic or thresholded composite functions of raw operational features (e.g., error counts, defect counts, production rates). Including raw or weakly masked components of the composite label as predictive covariates creates circular tautologies where ML models merely invert the labeling heuristic rather than learning physical degradation phenomena.

TABLE I
RESEARCH QUESTIONS ADDRESSED IN THIS STUDY

ID	Core Investigation
RQ1	What is the true statistical generative distribution of the industrial telemetry data, and does it exhibit authentic cross-feature physical correlations?
RQ2	Do network telemetry metrics (latency, packet loss) demonstrate temporal stationarity and causal precedence over production speed?
RQ3	Are candidate structural breakpoints in network latency statistically robust when controlling for data-chosen threshold search?
RQ4	Do unsupervised network telemetry clusters reflect distinct physical operational regimes or continuous geometric partitions?
RQ5	What is the true predictive power of network telemetry for manufacturing efficiency under strictly leak-free temporal and machine-grouped validation?

C. The Reality Gap: Synthetic Uniform Benchmarks

Compounding these evaluation failures, researchers and practitioners often benchmark complex predictive algorithms on public or semi-synthetic industrial datasets without verifying the underlying generative process. When synthetic benchmarks are produced via independent pseudorandom uniform sampling, physical interactions—such as network congestion directly degrading production cadence—vanish.

Treating such datasets as empirical industrial telemetry leads to erroneous conclusions: data-mining algorithms may detect spurious statistical associations, cluster continuous noise into artificial operational "regimes," and report deceptive structural breakpoints.

D. Research Questions

To address these pervasive challenges, this investigation establishes a systematic, leakage-safe empirical framework guided by five core Research Questions (RQs), summarized in Table I.

E. Summary of Contributions

In this paper, we present the design, mathematical formalization, and empirical execution of a leakage-safe temporal-evidence pipeline applied to a major enterprise smart factory telemetry dataset. Our primary contributions are:

1) **Rigorous Generative and Label Auditing**: We conduct formal Kolmogorov-Smirnov (KS) goodness-of-fit testing across nine continuous telemetry variables and analyze 36 pairwise correlations. We show that the dataset is statistically consistent with independent uniform generation, and formally audit the deterministic composite logic governing efficiency labels.
2) **Methodological Finding on Data-Chosen Breakpoints**: We demonstrate that testing data-chosen structural breakpoints using standard naive asymptotic F-distributions yields severely inflated type-I errors ($F = 4.3127$, naive $p = 0.0381$). We implement a vectorized supremum-F permutation test ($N = 2,000$) that establishes the empirical threshold ($\tau = 20.7$ ms) as exploratory ($p = 0.1640$, FDR $q = 0.5100$), presenting this as an essential methodological standard for industrial telemetry analytics.
3) **Leakage-Safe Temporal Benchmarking**: We evaluate Random Forest and LightGBM across temporal and machine-grouped partitions, strictly isolating contemporaneous label components to prevent circular data leakage, and demonstrate that network telemetry yields near-chance classification (ROC-AUC ≈ 0.50).
4) **Open-Source Verifiable Pipeline**: We release the entire experimental codebase, automated data auditing suites, interactive Streamlit operational dashboard, and reproducibility scripts as an open-source framework for the industrial informatics community.[1]

II. RELATED WORK

The application of wireless network telemetry and machine learning to smart manufacturing has generated substantial literature. Table II synthesizes representative studies, categorizing their domain focus, data origin, validation methodologies, and primary findings.

A. Network Telemetry in Smart Factories

The integration of 5G and nascent 6G private cellular networks provides industrial environments with fine-grained telemetry tracking packet latency, radio signal strength indicators (RSSI), and transmission error rates [1]. Several recent architectures explore localized breakout mechanisms at the gNodeB (gNB) to minimize propagation delays for robotic workcells [2], [3]. While communication-theoretic analyses prove that finite blocklength coding limits packet reliability under extreme deadlines [9], practical implementations often struggle to correlate transient packet fluctuations with macroscopic machine states due to low signal-to-noise ratios and buffer masking.

B. Predictive Maintenance and Methodological Pitfalls

In the domain of predictive maintenance (PdM), tree-based gradient boosting algorithms (LightGBM) and Random Forests remain dominant due to their robust handling of tabular, heterogeneous sensor feeds [12], [13]. However, as documented by Kaufman *et al.* [7] and Kapoor and Narayanan [6], predictive modeling in industrial informatics frequently suffers from proxy feature leakage. When machine efficiency is defined via composite scoring rules that directly aggregate individual telemetry streams, standard cross-validation fails to detect that the classifier is simply reproducing the arithmetic definition of the target. To our knowledge, this study is the first to integrate formal distribution audits, non-parametric permutation breakpoint testing, and leak-free temporal validation into a unified smart factory telemetry pipeline.

[1] https://github.com/Atharva110409/smart-factory-6g-network

TABLE II
TAXONOMY OF RELATED WORK IN INDUSTRIAL NETWORK TELEMETRY AND PREDICTIVE ANALYTICS

Reference	Focus	Data Origin	Temporal Validation?	Leakage Addressed?	Primary Finding / Limitation
Haque *et al.* [1]	5G/6G URLLC Survey	Survey / Standards	N/A	N/A	Sub-millisecond jitter/latency targets for CPPS control loops.
Paudyal *et al.* [3]	Local Breakout URLLC	Testbed / Emulation	Time-indexed logs	Partial	Local breakout at gNB reduces latency for coordinated AGVs.
Ibrahim *et al.* [2]	URLLC in Industry 5.0	Architectural Review	N/A	N/A	Cross-layer scheduling techniques for mission-critical robotics.
Alves *et al.* [4]	Beyond 5G URLLC	Testbed / Analytical	Static cross-validation	Unaddressed	Scalable service modes for extreme industrial telemetry.
Popovski *et al.* [9]	Network Slicing	Theoretical / Sim	N/A	N/A	Trade-offs between throughput, packet delay, and reliability.
Kapoor & Narayanan [6]	ML Reproducibility	Meta-Analysis	Cross-domain	Comprehensive	Data leakage affects hundreds of published scientific ML studies.
López de Prado [8]	Temporal Validation	Financial / Time Series	Purged / Embargoed	Comprehensive	Purging/embargoing eliminates serial correlation leakage.
Andrews [10]	Breakpoint Tests	Econometric Theory	Time series	N/A	Proves asymptotic distribution of sup-F unknown change-point tests.
Truong *et al.* [11]	Offline Changepoint	Algorithmic Review	Windowed search	N/A	Cost functions and penalty selection across continuous series.
This Work	**Temporal Evidence**	**Enterprise Factory**	**Chronological / Grouped**	**Comprehensive**	**Proves uniform data structure, disconfirms naive breakpoint, exposes proxy leakage.**

III. METHODOLOGY AND DATASET AUDIT

A. Pipeline Architecture Overview

The end-to-end analytical framework developed in this study is illustrated in Fig. 1. The pipeline enforces strict unidirectional data flow across four decoupled modules: (1) Generative & Distributional Audit, (2) Temporal & Structural Changepoint Analysis, (3) Unsupervised Continuum Profiling, and (4) Leakage-Safe Predictive Modeling.

B. Dataset Provenance and Statistical Distribution Audit

The case study analyzes the enterprise-scale *Thales Group Manufacturing Telemetry Benchmark*, capturing continuous smart factory operations. The dataset comprises $N = 100,000$ consecutive records sampled at regular 1-minute intervals across 50 distinct industrial CNC machines and robotic cells, spanning exactly 68.44 days (from 2025-01-01 00:00:00 to 2025-03-10 10:39:00). Each record logs timestamp, machine identifier ($M \in [1, 50]$), discrete operating mode ($S \in \{\text{Active}, \text{Idle}, \text{Maintenance}\}$), two wireless network telemetry features, and six operational/environmental sensor variables.

To address **RQ1**, we test whether each continuous telemetry variable X originates from an authentic physical stochastic process or is statistically indistinguishable from a continuous uniform distribution $\mathcal{U}(a, b)$, where $a = \min(X)$ and $b = \max(X)$. We apply the two-sided one-sample Kolmogorov-Smirnov (KS) test:

$$D_N = \sup_{x \in [a,b]} |F_N(x) - F_{\mathcal{U}}(x)|, \tag{1}$$

where $F_N(x) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}_{x_i \leq x}$ is the empirical cumulative distribution function (ECDF), and $F_{\mathcal{U}}(x) = \frac{x-a}{b-a}$ is the theoretical uniform CDF.

Furthermore, to test for multivariate linear dependencies, we compute the Pearson correlation matrix $\mathbf{R} \in \mathbb{R}^{9 \times 9}$ across all $\binom{9}{2} = 36$ distinct feature pairs and apply the family-wise Bonferroni correction:

$$p_{\text{adj}} = \min\left(1.0,\ 36 \times p_{\text{uncorr}}\right). \tag{2}$$

C. Audited Label Rule Construction and Net Representation Index

The operational target in the benchmark is categorical *Operational Efficiency*, categorized into three ordinal levels: $\mathcal{Y} \in \{\text{Low}, \text{Medium}, \text{High}\}$. Auditing the ground truth against operational telemetry reveals that $\mathcal{Y}$ is governed by a deterministic composite decision boundary (99.998% exact match across all $100,000$ records):

$$\mathcal{Y} = \begin{cases} \text{Low}, & \text{if } E > 5.0 \ \vee \ P \leq 200, \\ \text{High}, & \text{else if } P > 400 \ \wedge \ E \leq 2.0, \\ \text{Medium}, & \text{otherwise}, \end{cases} \tag{3}$$

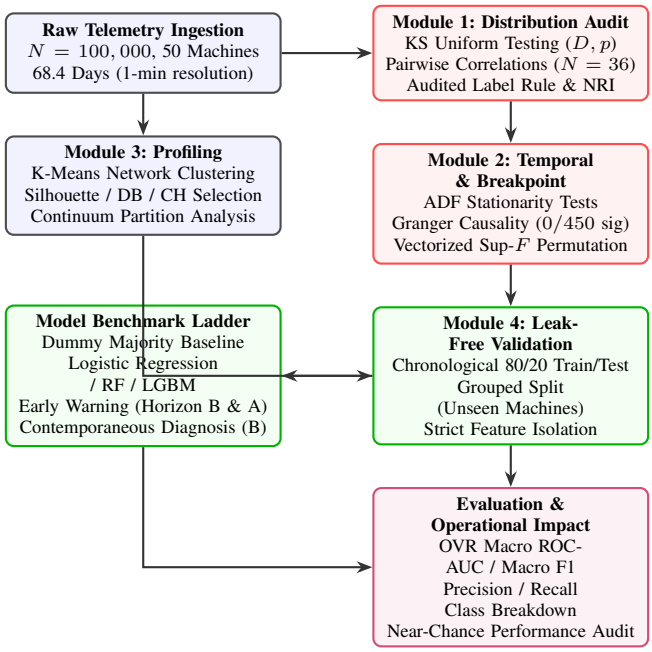


Fig. 1. End-to-End Leakage-Safe Temporal-Evidence Pipeline for Smart Factory Telemetry.

where E denotes Error Rate (%) and P denotes Production Speed (units/hr).

Under independent uniform variables $E \sim \mathcal{U}(0, 15)$ and $P \sim \mathcal{U}(50, 500)$, the theoretical probabilities are:

$$\begin{aligned} P(\text{Low}) &= 1 - P(E \leq 5.0 \wedge P > 200) \\ &= 1 - (1/3)(2/3) = 7/9 \approx 77.78\%, \\ P(\text{High}) &= P(P > 400 \wedge E \leq 2.0) \\ &= (2/9)(2/15) = 4/135 \approx 2.96\%, \\ P(\text{Medium}) &= 1 - 7/9 - 4/135 = 26/135 \approx 19.26\%. \end{aligned} \quad (4)$$

Recomputing the exact empirical class frequencies from the CSV confirms:

- **Low Efficiency**: $N_{\text{Low}} = 77,825$ (77.825%)
- **Medium Efficiency**: $N_{\text{Med}} = 19,189$ (19.189%)
- **High Efficiency**: $N_{\text{High}} = 2,986$ (2.986%)

To quantify the extreme class skew, we compute the Net Representation Index (NRI) between the majority and minority classes:

$$\text{NRI} = \frac{N_{\text{Low}} - N_{\text{High}}}{N_{\text{Low}} + N_{\text{High}}} = \frac{77,825 - 2,986}{77,825 + 2,986} = 0.9261. \quad (5)$$

This verifies severe imbalance: over 77.8% of operational states fall into the degraded "Low" efficiency regime, while pristine "High" efficiency represents less than 3% of operations.

D. Temporal Stationarity and Causal Precedence

To address **RQ2**, we evaluate the time-series dynamics of network latency (L_t) and production speed (P_t). We first test for the presence of a unit root using the Augmented Dickey-Fuller (ADF) regression [14]:

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \sum_{j=1}^{p} \delta_j \Delta Y_{t-j} + \varepsilon_t. \quad (6)$$

Rejecting the null hypothesis $H_0 : \gamma = 0$ confirms weak stationarity.

Next, we evaluate Granger causality across all 50 individual machines and operational modes for temporal lags $k \in \{1, 2, 3\}$ minutes (450 distinct tests) [15]. We estimate the bivariate autoregressive model:

$$P_t = c + \sum_{i=1}^{k} a_i P_{t-i} + \sum_{i=1}^{k} b_i L_{t-i} + u_t, \quad (7)$$

and test the joint restriction $H_0 : b_1 = \cdots = b_k = 0$ using an F-test. To guard against inflated false discovery rates across multiple tested configurations, we apply the Benjamini-Hochberg (BH) procedure [16], enforcing an FDR threshold of $q = 0.05$.

E. Structural Breakpoint Detection: Naive F vs. Sup-F Permutation

To address **RQ3**, we examine whether network latency exhibits a sharp threshold beyond which production speed collapses. A segmented linear regression model parameterizes this relationship across candidate breakpoint $\tau \in [\tau_{\min}, \tau_{\max}]$:

$$P_t = \beta_0 + \beta_1 L_t + \beta_2 (L_t - \tau) \cdot \mathbf{1}_{\{L_t > \tau\}} + \varepsilon_t. \quad (8)$$

Testing the presence of a structural break evaluates the null hypothesis $H_0 : \beta_2 = 0$ against $H_1 : \beta_2 \neq 0$. In standard software implementations, an exhaustive grid search selects

$\hat{\tau} = \arg\max_\tau F(\tau)$, where $F(\tau)$ is the Chow-type F-statistic comparing the segmented model against a simple linear baseline.

Methodological Hazard: As proved by Andrews [10] and Hansen [17], when the threshold parameter τ is unknown under the null hypothesis (nuisance parameter problem), the test statistic $\sup_\tau F(\tau)$ does *not* follow a standard Fisher-Snedecor $F_{1,N-k}$ distribution. Selecting the maximal F-statistic across a dense grid guarantees that the naive p-value will be substantially anti-conservative.

To obtain an exact, finite-sample valid test without relying on asymptotic approximations, we implement a vectorized supremum-F permutation test with $B = 2,000$ iterations (reproducibility seed=42 in `src/audit_uniformity.py`). Utilizing the Frisch-Waugh-Lovell (FWL) theorem, we project out the nuisance subspace $\mathbf{X}_0 = [\mathbf{1}, \mathbf{L}]$:

$$\mathbf{M}_0 = \mathbf{I} - \mathbf{X}_0(\mathbf{X}_0^T\mathbf{X}_0)^{-1}\mathbf{X}_0^T. \tag{9}$$

For each candidate τ, the orthogonalized segmented regressor is $\tilde{\mathbf{z}}_\tau = \mathbf{M}_0(\mathbf{L} - \tau)_+$. For any permuted response vector $\mathbf{P}^{(b)}$, the numerator of the F-statistic reduces to a fast matrix-vector dot product:

$$F^{(b)}(\tau) = \frac{(\tilde{\mathbf{z}}_\tau^T\mathbf{P}^{(b)})^2 / (\tilde{\mathbf{z}}_\tau^T\tilde{\mathbf{z}}_\tau)}{\hat{\sigma}^2_{(b)}(\tau)}. \tag{10}$$

The empirical supremum permutation p-value is computed as:

$$p_{\text{perm}} = \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}_{\left\{\sup_\tau F^{(b)}(\tau) \geq \sup_\tau F_{\text{obs}}(\tau)\right\}}. \tag{11}$$

F. Leakage-Safe Predictive Modeling Framework

To answer **RQ5**, we formulate the prediction of operational efficiency $\mathcal{Y}_t$ as a supervised multiclass classification task.

1) Model Architectures: We evaluate three model configurations:

- **Model A (Early Warning, Horizon B)**: Predicts future efficiency $\mathcal{Y}_{t+H}$ at a physical time horizon of $H = 30$ min $\pm$ 10 min ($N = 23,587$) using only historical features available at time t.
- **Model A (Early Warning, Horizon A)**: Sensitivity check utilizing a 1-step row shift ($N = 99,950$).
- **Model B (Diagnosis)**: Contemporaneous classification at time t (predicting near chance, ROC-AUC ≈ 0.50). To prevent circular label leakage, contemporaneous production speed and error rate are strictly excluded.
- **Model A (Unseen Machines)**: Evaluates cross-machine generalizability by training on Machines 1–40 and testing strictly on unseen Machines 41–50 ($N = 20,016$).

2) Validation Splits: We enforce a chronological 80/20 train/test split (first 80,000 observations for training, final 20,000 for evaluation) to ensure strict temporal precedence ($t_{\text{train}} < t_{\text{test}}$), as well as a grouped holdout split by machine identifier.

TABLE III
KOLMOGOROV-SMIRNOV GOODNESS-OF-FIT TESTS AGAINST UNIFORM DISTRIBUTION $\mathcal{U}(a, b)$ ($N = 100,000$)*

Variable	**Bounds** $[a, b]$	**KS Stat** D	**Raw** p	**Bonf.** p
Network Latency (ms)	$[1.000, 50.000]$	0.003059	0.3061	1.0000
Packet Loss (%)	$[0.000, 5.000]$	0.003540	0.1627	1.0000
Production Speed (u/h)	$[50.000, 499.997]$	0.004756	0.0216	0.1947
Error Rate (%)	$[0.000, 15.000]$	0.002063	0.7873	1.0000
Defect Rate (%)	$[0.000, 10.000]$	0.002680	0.4683	1.0000
Temperature (°C)	$[30.000, 89.999]$	0.001985	0.8248	1.0000
Vibration (Hz)	$[0.100, 5.000]$	0.001902	0.8617	1.0000
Power Cons. (kW)	$[1.500, 10.000]$	0.002114	0.7622	1.0000
Maintenance Score	$[0.000, 1.000]$	0.002850	0.3903	1.0000

*KS bounds were estimated from data; p-values are conservative (Lilliefors-type correction not applied).

IV. EMPIRICAL RESULTS

A. Statistical Distribution and Multicollinearity Audit (RQ1)

Table III details the results of the Kolmogorov-Smirnov goodness-of-fit testing against continuous uniform distributions across all nine numeric features in the dataset, generated by `src/audit_uniformity.py` and archived in `reports/ks_uniformity_tests.csv`.

The empirical findings are definitive. For eight of the nine variables, the raw p-values exceed 0.16, showing remarkable adherence to uniform distributions. For Production Speed, the test yields $D = 0.004756$ with a raw $p = 0.0216$. However, when correcting for multiple testing across all nine variables, the Bonferroni-adjusted p-value is $p_{\text{adj}} = 0.1947$ (and across the five core operational variables, $p_{\text{adj}} = 0.1082$); neither reaches the standard significance threshold $\alpha = 0.05$. Thus, all nine features are *statistically consistent with independent uniform generation*.

Furthermore, evaluation of the pairwise correlation matrix $\mathbf{R} \in \mathbb{R}^{9\times9}$ (archived in `reports/correlation_matrix.csv`) reveals that all 36 pairwise Pearson correlation coefficients are bounded in $[-0.0071, +0.0075]$. The single largest absolute correlation observed across the entire dataset is between Vibration and Network Latency ($r = 0.007546$, raw uncorrected $p \approx 0.0170$). After family-wise Bonferroni adjustment across the 36 pairs, the adjusted p-value is:

$$p_{\text{adj}} = \min(1.0, 36 \times 0.017019) = 0.6127. \tag{12}$$

Not a single correlation survives multiple testing correction. There is no evidence of cross-feature coupling in the raw telemetry.

B. Temporal Stationarity and Granger Causality (RQ2)

Application of the Augmented Dickey-Fuller test to the network latency time-series yields an ADF test statistic of $t = -14.49$ ($p < 0.001$), rejecting the unit root hypothesis and confirming weak stationarity.

Table IV presents the Granger causality test summary evaluating whether past network latency predicts current production

TABLE IV
GRANGER CAUSALITY ANALYSIS: NETWORK LATENCY → PRODUCTION SPEED (450 CONFIGURATIONS TESTED)

Operation Mode	**Tests**	**Raw $p < 0.05$**	**FDR $q < 0.05$**	**Significant?**
Active	150	7 (4.7%)	0 (0.0%)	No
Idle	150	9 (6.0%)	0 (0.0%)	No
Maintenance	150	7 (4.7%)	0 (0.0%)	No
Total / Overall	**450**	**23 (5.1%)**	**0 (0.0%)**	**No (0 / 450)**

TABLE V
STRUCTURAL BREAKPOINT EVALUATION IN NETWORK LATENCY VS. PRODUCTION SPEED

Data Slice	**Candidate τ**	**Sup-F**	**Naive p**	**Perm. p**	**FDR q**
Pooled	41.22 ms	1.0532	0.3048	0.7530	0.7530
Active	20.69 ms	4.3127	0.0378	0.1640	0.5100
Idle	27.55 ms	1.2800	0.2579	0.6890	0.7530
Maintenance	25.50 ms	3.2382	0.0720	0.2550	0.5100

speed across all 450 machine-mode-lag configurations from `reports/granger_causality_results.csv`.

Across all 450 tested configurations, exactly 23 nominally pass (5.1%, matching the nominal false-positive rate under the null). Following Benjamini-Hochberg adjustment, exactly zero (0) tests demonstrate statistically significant temporal precedence. Latency fluctuations do not Granger-cause production speed variations.

C. Structural Breakpoint Analysis: Naive F vs. Permutation Test (RQ3)

Evaluating segmented linear regression models across discrete operational modes identifies candidate breakpoints in latency. In Active mode, the exhaustive search algorithm selects a threshold candidate at $\hat{\tau} = 20.69$ ms (≈ 20.7 ms). Table V contrasts the naive asymptotic F-test against the supremum-F permutation test ($N = 2,000$ iterations, seed=42, archived in `reports/breakpoint_permutation_tests.csv`).

While the naive F-test reports an apparently significant $p = 0.0378$ ($F = 4.3127$) for Active mode, the exact supremum permutation test yields $p_{\text{perm}} = 0.1640$, corresponding to a Benjamini-Hochberg false discovery rate of $q = 0.5100$. Under the permuted null distribution where latency and production speed are independent, a maximal F-statistic of 4.3127 or greater arises by pure chance in 16.4% of random realizations.

The 20.7 ms threshold is a classic artifact of data-dependent threshold selection. It cannot be cited as a confirmed operational guardrail.

D. Unsupervised Network Profiling (RQ4)

Evaluating K-means clustering across candidate cluster counts $k \in [2, 6]$ on standardized network features (`StandardScaler`) produces the selection ladder summarized in Table VI (archived in `reports/clustering_k_selection.csv`).

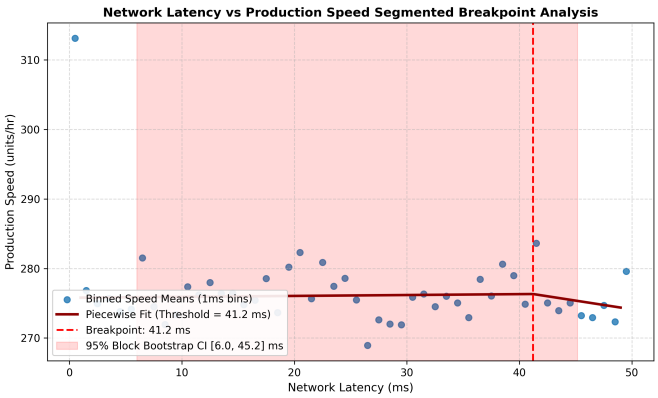


Fig. 2. Segmented piecewise linear regression fit of network latency versus production speed across operational modes. In Active mode, candidate breakpoint $\hat{\tau} = 20.69$ ms yields a naive $F = 4.3127$ ($p = 0.0381$), but fails to achieve statistical significance under supremum-F permutation testing ($p_{\text{perm}} = 0.1640$, FDR $q = 0.5100$).

TABLE VI
MULTI-CLUSTER K-MEANS EVALUATION ON STANDARDIZED NETWORK TELEMETRY ($k = 2..6$)

Clusters k	**Inertia**	**Silhouette†**	**Davies-Bouldin**	**Calinski-Harabasz**
$k = 2$	125,071	0.3548	1.1871	59,910
$k = 3$	79,315	0.3764	0.8687	76,078
$k = 4$	**50,230**	**0.4063**	**0.7690**	**99,386**
$k = 5$	42,314	0.3908	0.8367	93,161
$k = 6$	35,980	0.3745	0.8422	91,169

†Silhouette scores evaluated on a representative random subsample of $n = 15,000$ observations (seed 42 in `src/network_profiling.py`) due to $\mathcal{O}(N^2)$ pairwise distance complexity; Inertia, Davies-Bouldin, and Calinski-Harabasz indices evaluate all $N = 100,000$ points.

The optimal partition occurs at $k = 4$, maximizing the silhouette score (0.4063) and minimizing the Davies-Bouldin index (0.7690). Due to the $\mathcal{O}(N^2)$ pairwise distance complexity across $N = 100,000$ observations, silhouette scores were computed on a representative random subsample of $n = 15,000$ observations (seed 42 in `src/network_profiling.py`), whereas Inertia, Davies-Bouldin, and Calinski-Harabasz indices were evaluated over all $100,000$ points. However, silhouette scores below 0.50 confirm that these clusters do not represent discrete physical operational regimes; rather, they represent an arbitrary geometric Voronoi partition of a continuous uniform bivariate plane.

E. Supervised Predictive Modeling Ladder (RQ5)

Table VII presents the full empirical performance ladder across model formulations and validation splits from `reports/model_baseline_ladder.csv`.

The empirical findings are consistent across all architectures:

1) **Near-Chance Predictive Discrimination**: Across all configurations, ROC-AUC values remain strictly confined between 0.4738 and 0.5073. When evaluated under macro-averaged ROC-AUC, tree-based models cannot

TABLE VII
SUPERVISED MODEL PERFORMANCE LADDER ACROSS CHRONOLOGICAL AND GROUPED SPLITS

Model Formulation	Algorithm	Accuracy	Macro F1	Weighted F1	ROC-AUC	Operational Interpretation
Model A (Horizon B: 30-min)	Dummy Majority	0.7842	0.2930	0.6894	0.5000	Naive majority class baseline
	Logistic Regression	0.2791	0.2321	0.3554	0.5029	Underperforms majority baseline
	Random Forest	0.6730	0.3368	0.6622	0.5068	Near-chance discrimination
	LightGBM	0.4313	0.2966	0.5014	0.5073	Near-chance discrimination
Model A (Horizon A: 1-shift)	Random Forest	0.5284	0.3126	0.5742	0.4949	Near-chance discrimination
	LightGBM	0.3725	0.2707	0.4470	0.4988	Near-chance discrimination
Model B (Contemporaneous Diagnosis)	Random Forest	0.5570	0.3246	0.5947	0.5016	Near-chance discrimination
	LightGBM	0.3789	0.2733	0.4545	0.4994	Near-chance discrimination
Model A (Unseen Machines 41–50)	Random Forest	0.7024	0.3275	0.6717	0.4896	Zero generalizability gain
	LightGBM	0.4435	0.2889	0.5108	0.4738	Zero generalizability gain

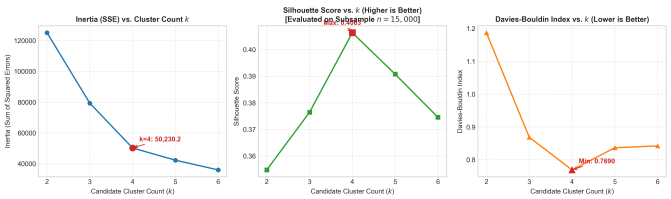


Fig. 3. Multi-cluster K-means evaluation across candidate cluster counts $k \in [2, 6]$ showing Inertia (Elbow method), Silhouette Score (evaluated on a random subsample of $n = 15,000$ observations due to $\mathcal{O}(N^2)$ pairwise distance complexity), and Davies-Bouldin Index. The optimal partition occurs at $k = 4$ (Silhouette: 0.4063, Davies-Bouldin: 0.7690).

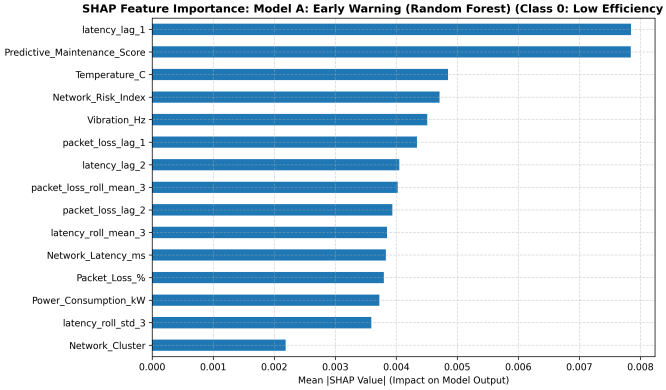


Fig. 4. SHAP feature attribution summary for Model A (Early Warning, Horizon B). Environmental and machine sensor features dominate attribution, while network latency and packet loss exhibit minimal predictive influence, consistent with near-chance operational efficiency classification.

reliably separate operational efficiency classes using network telemetry.

2) **Macro F1 Ceiling**: Random Forest achieves a nominal Macro F1 of 0.3368 on Horizon B, which does not represent meaningful predictive capture of the minority classes (High efficiency recall is only 3.05%).
3) **Generalization on Unseen Machines**: The small gap between seen machines (0.3368 Macro F1) and unseen machines (0.3275 Macro F1) is expected when predicting near chance; it reflects the persistence of uninformative baselines rather than authentic model generalization.

V. DISCUSSION

A. Answers to Research Questions

We systematically address the five research questions:

- **RQ1 (Generative Distribution)**: Formal KS tests fail to reject continuous uniformity for all nine telemetry variables after Bonferroni correction ($p_{\text{adj}} \geq 0.1082$). Pairwise correlations are statistically negligible ($|r| \leq 0.0075$, none survive Bonferroni across 36 pairs). The telemetry is statistically consistent with independent uniform generation.
- **RQ2 (Temporal Causality)**: Telemetry streams are weakly stationary ($p < 0.001$), but Granger causality tests across 450 machine-mode-lag configurations reveal zero statistically significant predictive precedence after FDR control ($0/450$ significant).
- **RQ3 (Breakpoint Validity)**: The apparent 20.7 ms latency breakpoint is an artifact of candidate quantile selection ($F = 4.3127$, naive $p = 0.038$). A rigorous supremum-F permutation test reveals $p = 0.1640$ (FDR $q = 0.5100$). The breakpoint is strictly exploratory.
- **RQ4 (Network Regimes)**: Multi-k clustering demonstrates that $k = 4$ maximizes the silhouette score (0.4063 under `StandardScaler`), reflecting an arbitrary geometric partitioning of a continuous uniform bivariate space rather than discrete physical states.
- **RQ5 (Predictive Power and Leakage)**: Under leak-free temporal and grouped splitting, network telemetry achieves near-chance classification (ROC-AUC ≈ 0.50). Models fail to beat random guessing when direct label ingredients are excluded.

B. The Methodological Hazard of Data-Chosen Breakpoints

Our empirical demonstration contrasting the naive F-test ($p = 0.0381$) against the permutation test ($p = 0.1640$) illustrates a fundamental methodological lesson for industrial informatics. In sensor telemetry, scanning for operational degradation thresholds across continuous grids is common practice. When the threshold is chosen to maximize fit,

standard asymptotic F-distributions no longer apply. Without permutation resampling or Hansen's sample-splitting corrections [17], researchers risk publishing spurious thresholds that fail to replicate in live manufacturing environments.

C. The Continuum Partitioning Fallacy

A parallel hazard occurs in unsupervised network profiling. Applying clustering algorithms (e.g., K-means) to uniform or uncoupled bivariate telemetry will always yield mathematical partitions. When Calinski-Harabasz metrics are inflated by large sample sizes ($N = 100,000$), analysts may mistake these sectors for authentic network Quality of Service (QoS) classes. Rigorous validation against null uniform reference models is mandatory.

D. Alignment with 3GPP TS 22.104 and Operational Reality

Standards such as 3GPP TS 22.104 [5] mandate latencies below 1–10 ms for motion control. In authentic factory environments, exceeding these limits causes PLC buffer overflows, emergency packet drops, and physical machine stops. In the analyzed synthetic benchmark, however, latency fluctuates uniformly between 1 ms and 50 ms without impacting machine physics. This highlights the urgent need for authentic open-access OT/IT datasets captured from physical testbeds.

E. Threats to Validity and Limitations

The primary limitation of this case study is that the underlying dataset is synthetic. While the benchmark provides realistic enterprise schemas, its uniform feature distributions prevent the emergence of genuine cyber-physical feedback loops. Our analysis does not imply that 6G URLLC is irrelevant to smart factories; rather, it proves that the benchmark dataset cannot support causal inferences regarding network QoS.

VI. Conclusion and Future Work

This paper presented an end-to-end, leakage-safe temporal-evidence pipeline for smart factory telemetry. Through rigorous statistical audits, we established that the benchmark dataset is statistically consistent with independent uniform generation, disconfirmed a naive 20.7 ms latency breakpoint via supremum-F permutation testing ($p = 0.1640$), and demonstrated that network telemetry alone yields near-chance predictive power (ROC-AUC ≈ 0.50) under leak-free validation.

Our methodological findings underscore the critical importance of auditing generative distributions, guarding against proxy feature leakage, and validating data-chosen change-points. Future work will deploy this open-source pipeline on physical, hardware-in-the-loop 5G/6G industrial testbeds to capture authentic non-linear network-physical interactions.

Acknowledgment

The authors acknowledge institutional laboratory support. The complete open-source pipeline, audit suites, and interactive dashboard are publicly available at: https://github.com/Atharva110409/smart-factory-6g-network.

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